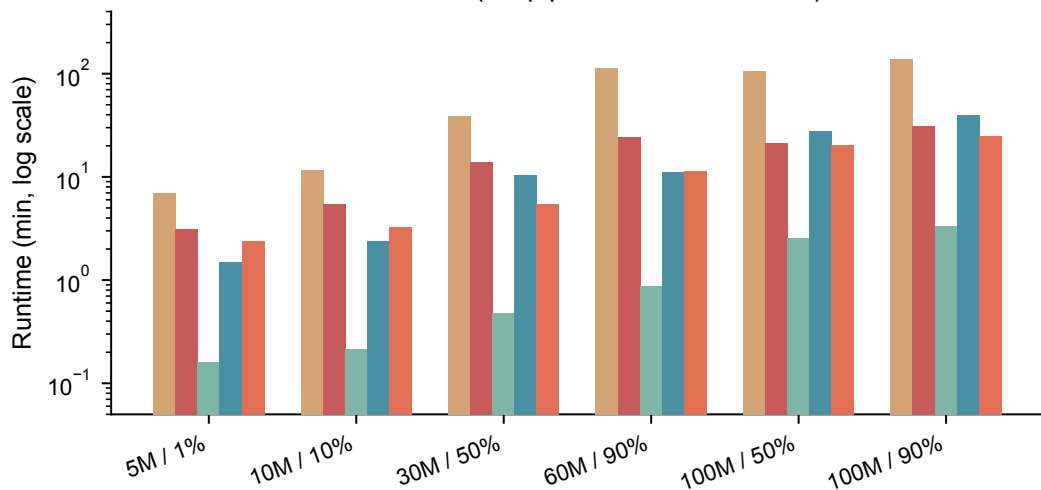
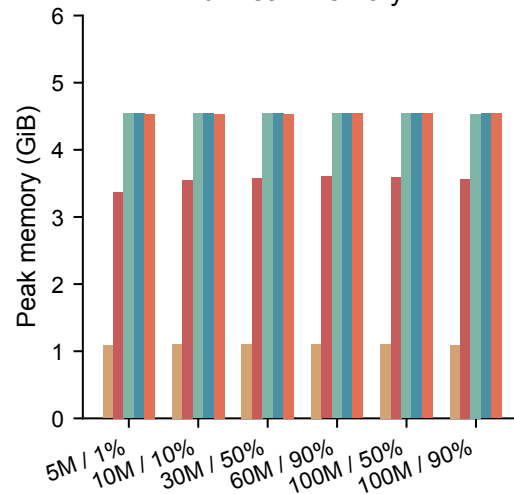


■ KneadData    
 ■ Hostile + fastp    
 ■ Deacon (depletion only)    
 ■ RustyClean AUTO (fastp + deacon + conditional Bowtie2 verification)    
 ■ RustyClean AUTO without recheck (fastp + deacon)

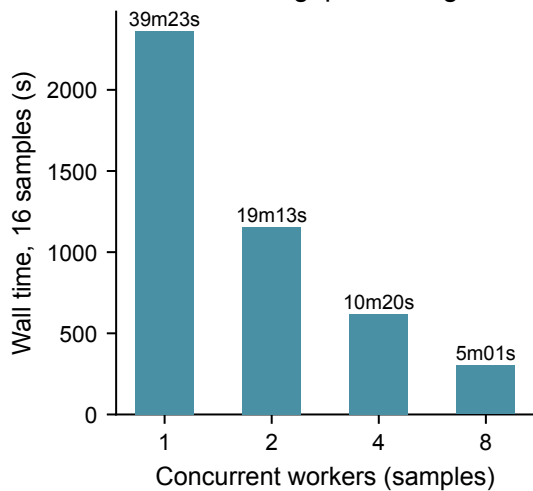
a Runtime (full pipeline or stated basis)



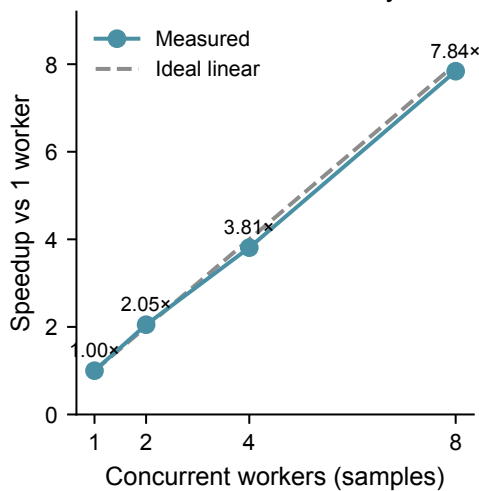
b Peak memory



c Throughput scaling



d Parallel efficiency



e Memory per worker stays flat

